

Full Analysis: Seasonality and Geography of Airport Flight Traffic

Kaden Zhu

June 04, 2026

Contents

1. Load and combine the data	1
2. Data quality check	2
3. Exploratory data analysis	3
3.1 Outcome distribution	3
3.2 Overall and grouped summaries	4
3.3 Seasonal pattern	6
3.4 Country comparison	7
3.5 The airport-level confound made visible	8
4. Regression models	9
4.1 Model 1, without airport control	9
4.2 Model 2, with airport fixed effects	10
4.3 Comparing the two models	11
4.4 Diagnostics	12
4.5 Log-scale robustness check	13
5. Interpretation in domain terms	14
6. Limitations	14
7. What I would do next	15

```
library(tidyverse)
library(scales)
library(broom)
library(knitr)
library(fixest)    # fast fixed-effects regression (install.packages("fixest"))
```

1. Load and combine the data

```
airport_2025 <- read_csv("data/airport_traffic_2025.csv")
airport_2026 <- read_csv("data/airport_traffic_2026.csv")

airport <- bind_rows(airport_2025, airport_2026) |>
  mutate(
    MONTH_MON = fct_reorder(MONTH_MON, MONTH_NUM),
    YEAR = factor(YEAR)
  )
```

```
dim(airport)
```

```
## [1] 125771    13
```

The combined frame holds every airport-day record from both years side by side. Converting MONTH_MON to an ordered factor keeps the calendar order on every plot and in the regression. Treating YEAR as a factor with two levels matches what the linear model needs.

2. Data quality check

```
airport |>
  summarize(across(everything(), ~ sum(is.na(.x)))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") |>
  arrange(desc(n_missing))
```

```
## # A tibble: 13 x 2
##   variable      n_missing
##   <chr>         <int>
## 1 FLT_DEP_IFR_2    89783
## 2 FLT_ARR_IFR_2    89783
## 3 FLT_TOT_IFR_2    89783
## 4 YEAR              0
## 5 MONTH_NUM         0
## 6 MONTH_MON         0
## 7 FLT_DATE          0
## 8 APT_ICAO          0
## 9 APT_NAME          0
## 10 STATE_NAME       0
## 11 FLT_DEP_1        0
## 12 FLT_ARR_1        0
## 13 FLT_TOT_1        0
```

```
tibble(
  n_records = nrow(airport),
  n_airports = n_distinct(airport$APT_ICAO),
  n_countries = n_distinct(airport$STATE_NAME),
  n_dates = n_distinct(airport$FLT_DATE)
)
```

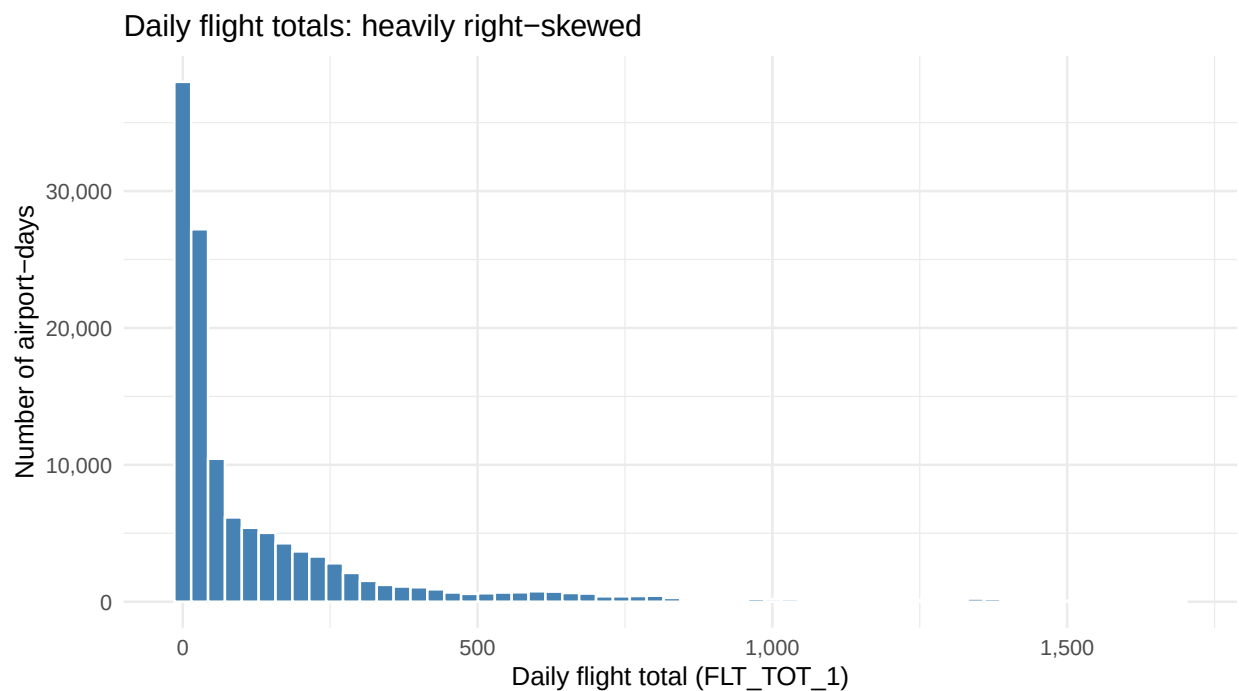
```
## # A tibble: 1 x 4
##   n_records n_airports n_countries n_dates
##   <int>     <int>     <int>    <int>
## 1   125771       332        42     396
```

I read missingness first because anything substantial in FLT_TOT_1 would change the interpretation of every summary statistic below. The record counts confirm the dataset's scale: hundreds of airports across a couple of years gives us a sample large enough to fit a regression with airport-level fixed effects without running out of degrees of freedom.

3. Exploratory data analysis

3.1 Outcome distribution

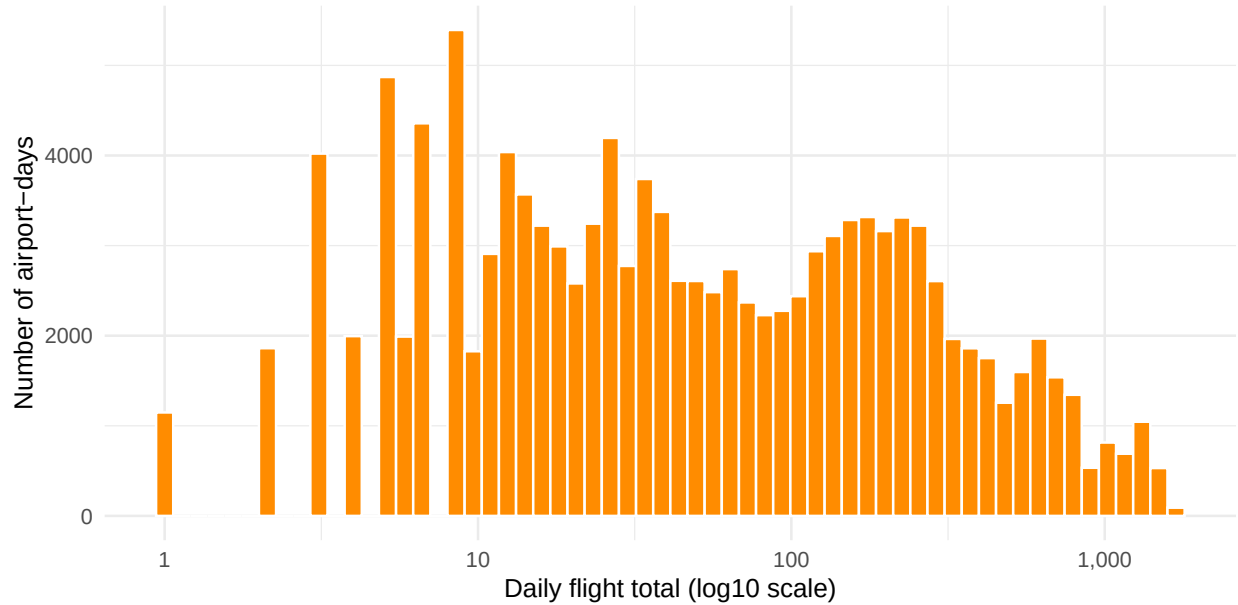
```
ggplot(airport, aes(x = FLT_TOT_1)) +  
  geom_histogram(bins = 60, fill = "steelblue", color = "white") +  
  scale_x_continuous(labels = comma) +  
  scale_y_continuous(labels = comma) +  
  labs(  
    title = "Daily flight totals: heavily right-skewed",  
    x = "Daily flight total (FLT_TOT_1)",  
    y = "Number of airport-days"  
  ) +  
  theme_minimal(base_size = 12)
```



```
ggplot(airport, aes(x = FLT_TOT_1 + 1)) +  
  geom_histogram(bins = 60, fill = "darkorange", color = "white") +  
  scale_x_log10(labels = comma) +  
  labs(  
    title = "Same distribution on a log10 scale",  
    subtitle = "Approximately symmetric, a useful sign for residual diagnostics",  
    x = "Daily flight total (log10 scale)",  
    y = "Number of airport-days"  
  ) +  
  theme_minimal(base_size = 12)
```

Same distribution on a log10 scale

Approximately symmetric, a useful sign for residual diagnostics



The raw histogram has the long right tail I expected from the boxplots in the proposal. On a log scale the distribution looks roughly symmetric, which tells me a log transformation of the outcome is a sensible thing to keep in reserve in case the regression residuals misbehave.

3.2 Overall and grouped summaries

```
airport |>
  summarize(
    n = n(),
    mean = mean(FLT_TOT_1, na.rm = TRUE),
    median = median(FLT_TOT_1, na.rm = TRUE),
    sd = sd(FLT_TOT_1, na.rm = TRUE),
    p10 = quantile(FLT_TOT_1, 0.10, na.rm = TRUE),
    p90 = quantile(FLT_TOT_1, 0.90, na.rm = TRUE),
    max = max(FLT_TOT_1, na.rm = TRUE)
  ) |>
  kable(digits = 1, caption = "Overall summary of daily flight totals.")
```

Table 1: Overall summary of daily flight totals.

n	mean	median	sd	p10	p90	max
125771	146	39	247.5	4	412	1688

The mean is well above the median and the 90th percentile is several times the median. This is the right tail in numbers rather than in a picture: the typical airport-day in this sample is much quieter than the average suggests.

```
airport |>
  group_by(MONTH_MON) |>
  summarize(
```

```

n = n(),
mean = mean(FLT_TOT_1, na.rm = TRUE),
median = median(FLT_TOT_1, na.rm = TRUE),
sd = sd(FLT_TOT_1, na.rm = TRUE),
.groups = "drop"
) |>
kable(digits = 1, caption = "Daily flight totals by month.")

```

Table 2: Daily flight totals by month.

MONTH_MON	n	mean	median	sd
JAN	19409	119.4	31	210.9
APR	9563	149.5	40	251.5
MAY	9901	157.1	44	259.6
JUL	9919	169.7	51	273.2
AUG	9879	170.0	51	275.1
SEP	9627	167.5	47	271.7
DEC	9821	130.1	32	229.4
FEB	8808	125.3	32	218.3
JUN	9594	166.6	49	269.3
MAR	9794	130.9	33	226.3
NOV	9564	130.7	32	230.2
OCT	9892	158.7	42	262.6

Across both years the monthly means rise from a winter low through spring, crest in mid-summer, and fall back through autumn. The medians track the means too. The seasonal pattern is not an artifact of the right tail.

```

top_countries <- airport |>
  group_by(STATE_NAME) |>
  summarize(mean_flights = mean(FLT_TOT_1, na.rm = TRUE), .groups = "drop") |>
  slice_max(mean_flights, n = 10) |>
  pull(STATE_NAME)

airport |>
  filter(STATE_NAME %in% top_countries) |>
  group_by(STATE_NAME) |>
  summarize(
    n = n(),
    n_airports = n_distinct(APT_ICAO),
    mean = mean(FLT_TOT_1, na.rm = TRUE),
    median = median(FLT_TOT_1, na.rm = TRUE),
    sd = sd(FLT_TOT_1, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(mean)) |>
  kable(digits = 1, caption = "Daily flight totals for the top 10 countries by mean.")

```

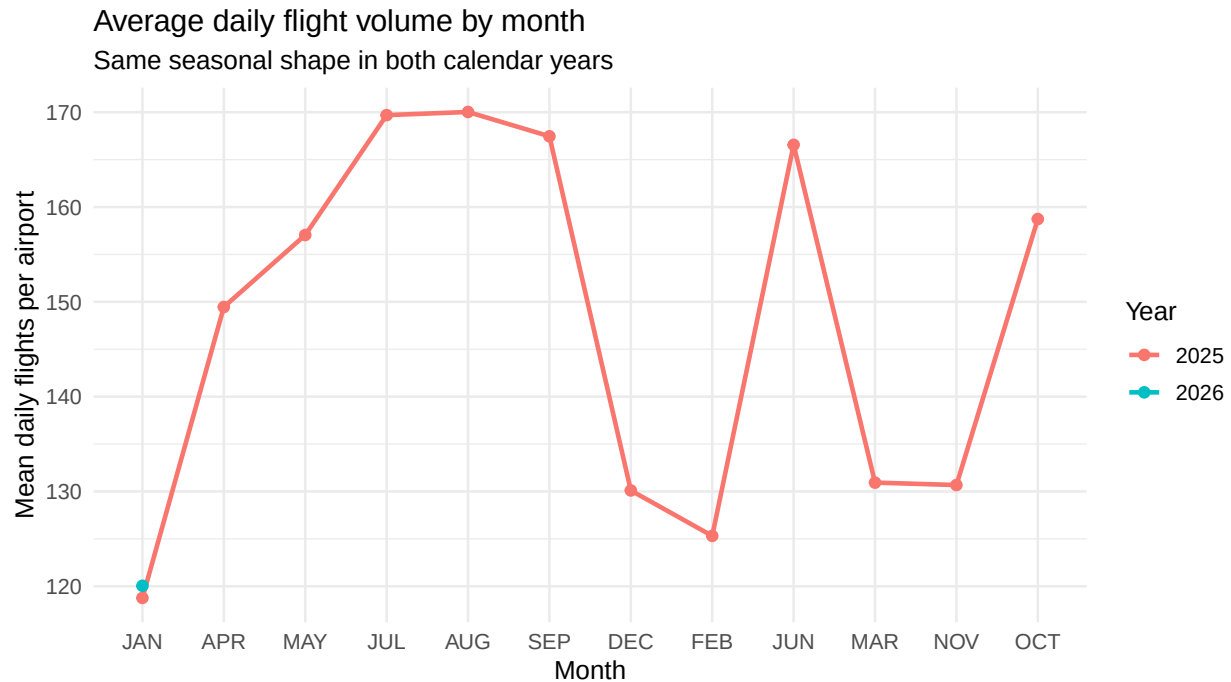
Table 3: Daily flight totals for the top 10 countries by mean.

STATE_NAME	n	n_airports	mean	median	sd
Denmark	396	1	695.6	709.5	103.4
Türkiye	2376	6	556.8	274.0	501.0
Finland	396	1	425.7	430.0	32.4
Switzerland	1187	3	406.7	476.0	295.4
Israel	396	1	384.5	394.5	118.4
Hungary	396	1	373.9	386.5	48.2
Germany	5932	15	305.8	177.0	348.9
Netherlands	1967	5	291.9	30.0	524.2
United Kingdom	7524	19	284.2	167.0	311.0
Italy	5544	14	261.3	200.0	223.5

The `n_airports` column is the key one in this table. The countries at the top of the ranking are not necessarily countries with many busy airports. Some have only one or two airports in the sample, and those airports happen to be very busy. That is exactly the confound the regression is built to address.

3.3 Seasonal pattern

```
airport |>
  group_by(MONTH_MON, YEAR) |>
  summarize(mean_flights = mean(FLT_TOT_1, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = MONTH_MON, y = mean_flights,
             group = YEAR, color = YEAR)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Average daily flight volume by month",
    subtitle = "Same seasonal shape in both calendar years",
    x = "Month",
    y = "Mean daily flights per airport",
    color = "Year"
  ) +
  theme_minimal(base_size = 12)
```

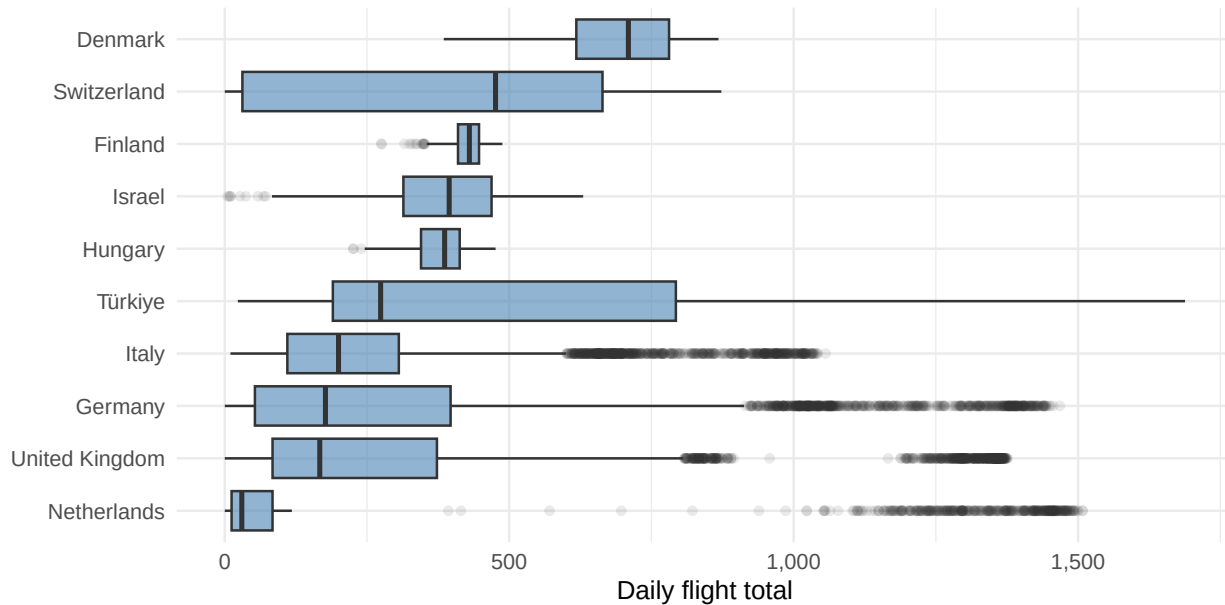


The two yearly lines move together across the calendar, so the seasonal effect is reproducible rather than year-specific. Where the lines diverge by a few flights per airport, that is the size of the year effect we will recover in the regression.

3.4 Country comparison

```
airport |>
  filter(STATE_NAME %in% top_countries) |>
  mutate(STATE_NAME = fct_reorder(STATE_NAME, FLT_TOT_1, .fun = median)) |>
  ggplot(aes(x = STATE_NAME, y = FLT_TOT_1)) +
  geom_boxplot(fill = "steelblue", alpha = 0.6, outlier.alpha = 0.1) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Daily flight totals by country (top 10 by mean)",
    subtitle = "Wide spreads come from hub-vs-regional variation within countries",
    x = NULL,
    y = "Daily flight total"
  ) +
  theme_minimal(base_size = 12)
```

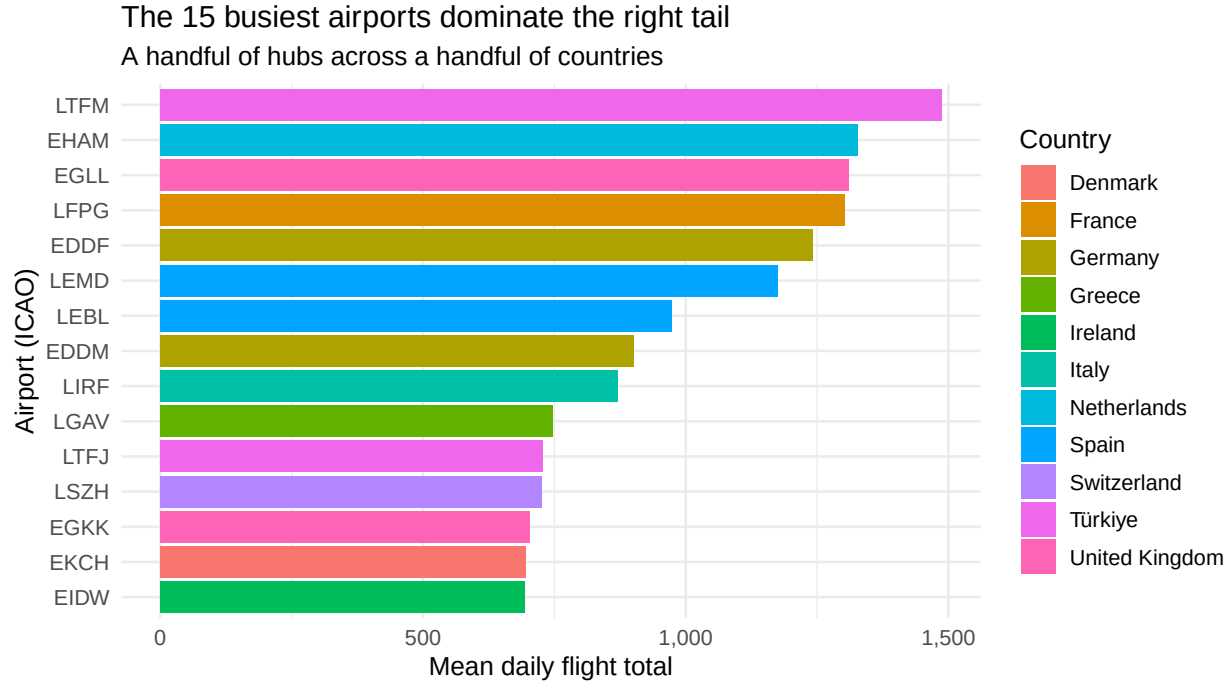
Daily flight totals by country (top 10 by mean)
Wide spreads come from hub-vs-regional variation within countries



The width of each box is a within-country story. Countries with both a major hub and several regional airports, such as Germany, the UK, and Italy, get the widest spreads because the same country produces both 1,500-flight days at the hub and 50-flight days at small airports.

3.5 The airport-level confound made visible

```
airport |>
  group_by(APT_ICAO, STATE_NAME) |>
  summarize(mean_flights = mean(FLT_TOT_1, na.rm = TRUE), .groups = "drop") |>
  slice_max(mean_flights, n = 15) |>
  mutate(APT_ICAO = fct_reorder(APT_ICAO, mean_flights)) |>
  ggplot(aes(x = APT_ICAO, y = mean_flights, fill = STATE_NAME)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  labs(
    title = "The 15 busiest airports dominate the right tail",
    subtitle = "A handful of hubs across a handful of countries",
    x = "Airport (ICAO)",
    y = "Mean daily flight total",
    fill = "Country"
  ) +
  theme_minimal(base_size = 12)
```



A small set of hubs sits far above the rest. When the boxplots above made a country look busy, this plot is usually the reason. The regression in section 4 puts these airports in as fixed effects so that month and year coefficients are estimated after holding airport identity constant.

4. Regression models

4.1 Model 1, without airport control

```
model1 <- lm(FLT_TOT_1 ~ MONTH_MON + STATE_NAME + YEAR, data = airport)

glance(model1) |>
  select(r.squared, adj.r.squared, sigma, statistic, p.value, df, nobs) |>
  kable(digits = 3, caption = "Model 1 fit statistics.")
```

Table 4: Model 1 fit statistics.

r.squared	adj.r.squared	sigma	statistic	p.value	df	nobs
0.199	0.199	221.565	588.909	0	53	125771

```
tidy(model1) |>
  filter(str_starts(term, "MONTH_MON") | term == "YEAR2026") |>
  kable(digits = 2, caption = "Model 1: month and year coefficients (intercept = January, baseline year 2025).")
```

Table 5: Model 1: month and year coefficients (intercept = January, baseline year 2025).

term	estimate	std.error	statistic	p.value
MONTH_MONAPR	31.34	3.20	9.81	0.00

term	estimate	std.error	statistic	p.value
MONTH_MONMAY	39.00	3.17	12.31	0.00
MONTH_MONJUL	51.66	3.17	16.31	0.00
MONTH_MONAUG	52.00	3.17	16.40	0.00
MONTH_MONSEP	49.60	3.19	15.54	0.00
MONTH_MONDEC	11.81	3.18	3.72	0.00
MONTH_MONFEB	6.83	3.26	2.09	0.04
MONTH_MONJUN	48.56	3.19	15.20	0.00
MONTH_MONMAR	12.45	3.18	3.92	0.00
MONTH_MONNOV	12.59	3.20	3.94	0.00
MONTH_MONOCT	40.64	3.17	12.82	0.00
YEAR2026	1.56	3.18	0.49	0.62

Model 1 is the naive picture. Its R^2 answers “how much of the variation in daily flights can month, country, and year jointly explain?” Its month coefficients give the average shift in daily flights for each month relative to January, ignoring the fact that the country mix doing those summer flights skews toward hub-heavy states.

4.2 Model 2, with airport fixed effects

```
# Absorb airport as a fixed effect instead of building 330 dummy
# columns, so the model fits quickly on the full data.
model2 <- feols(FLT_TOT_1 ~ MONTH_MON + YEAR | APT_ICAO, data = airport)

tibble(
  r.squared      = get_r2(model2),
  within.r.squared = fixest::r2(model2, type = "wr2"),
  n_airports_fe  = model2$fixef_sizes[["APT_ICAO"]],
  nobs           = nobs(model2)
) |>
  kable(digits = 3, caption = "Model 2 fit statistics (airport absorbed as fixed effect).")
```

Table 6: Model 2 fit statistics (airport absorbed as fixed effect).

r.squared	within.r.squared	n_airports_fe	nobs
0.965	0.16	332	125771

```
tidy_coefs(model2) |>
  filter(str_starts(term, "MONTH_MON") | term == "YEAR2026") |>
  kable(digits = 2, caption = "Model 2: month and year coefficients after controlling for airport.")
```

Table 7: Model 2: month and year coefficients after controlling for airport.

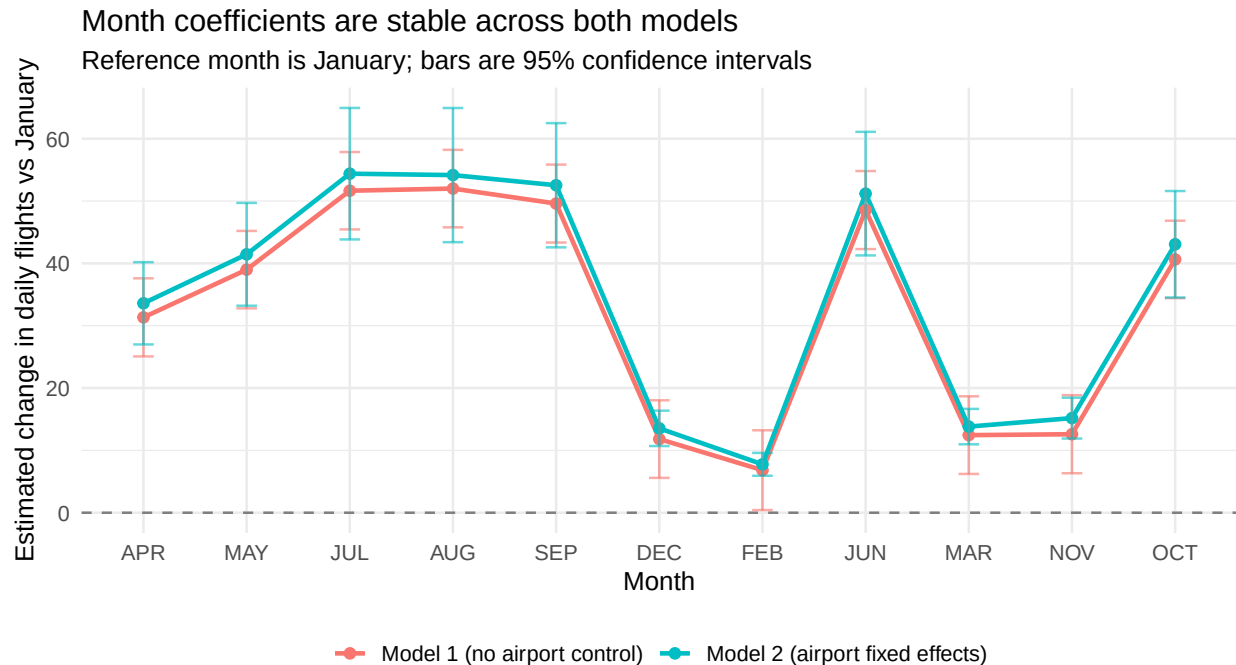
term	estimate	std.error	statistic	p.value
MONTH_MONAPR	33.60	3.37	9.98	0
MONTH_MONMAY	41.46	4.21	9.85	0
MONTH_MONJUL	54.39	5.38	10.11	0
MONTH_MONAUG	54.17	5.49	9.86	0
MONTH_MONSEP	52.54	5.08	10.34	0
MONTH_MONDEC	13.53	1.45	9.34	0

term	estimate	std.error	statistic	p.value
MONTH_MONFEB	7.76	0.94	8.29	0
MONTH_MONJUN	51.19	5.06	10.12	0
MONTH_MONMAR	13.81	1.45	9.54	0
MONTH_MONNOV	15.18	1.67	9.08	0
MONTH_MONOCT	43.07	4.36	9.88	0
YEAR2026	2.47	0.83	2.97	0

Model 2 is the model I actually want to interpret. Absorbing airport as a fixed effect soaks up the “which hubs are in the sample” effect, so the month coefficients here estimate the average within-airport seasonal effect: how much busier (or quieter) an airport is in a given month compared to its own January baseline. STATE_NAME is not in this model. Country is fully determined by airport, so including both would be perfectly collinear.

4.3 Comparing the two models

```
bind_rows(
  tidy_coefs(model1) |> mutate(model = "Model 1 (no airport control)"),
  tidy_coefs(model2) |> mutate(model = "Model 2 (airport fixed effects)")
) |>
  filter(str_starts(term, "MONTH_MON")) |>
  mutate(month = str_remove(term, "MONTH_MON"),
         month = fct_reorder(month, match(month, levels(airport$MONTH_MON)))) |>
  ggplot(aes(x = month, y = estimate, color = model, group = model)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = estimate - 1.96 * std.error,
                    ymax = estimate + 1.96 * std.error),
                width = 0.2, alpha = 0.6) +
  labs(
    title = "Month coefficients are stable across both models",
    subtitle = "Reference month is January; bars are 95% confidence intervals",
    x = "Month", y = "Estimated change in daily flights vs January",
    color = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```

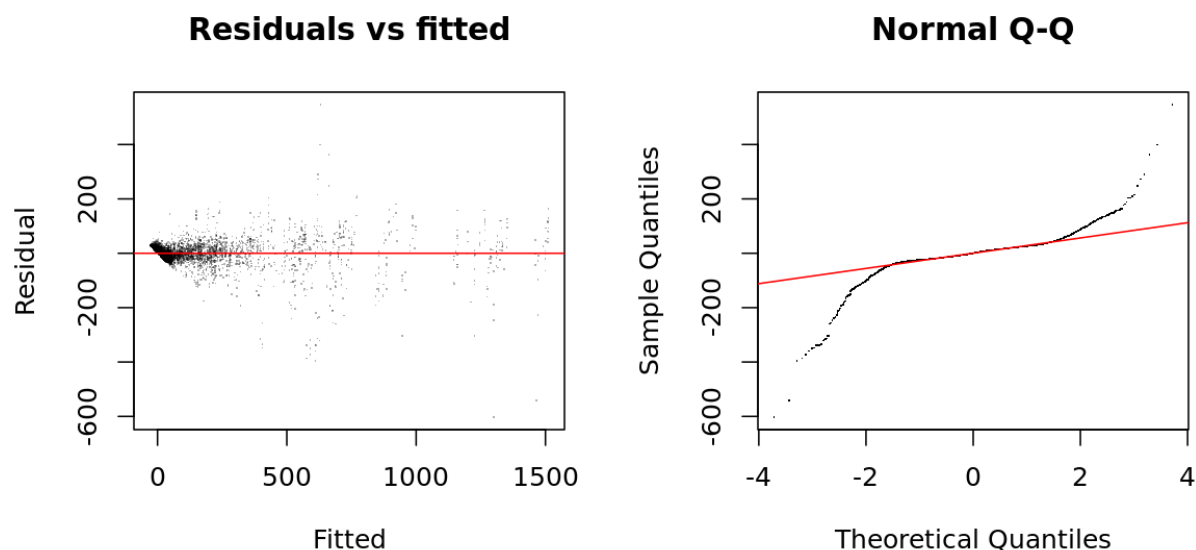


If the seasonal pattern were really a story about hub composition, the teal line (Model 2) would flatten. The closeness of the two lines shows the monthly effect is within-airport seasonal variation, not the byproduct of which countries happen to have records that month.

4.4 Diagnostics

```
# plot(model2) would compute leverage for every observation, slow at
# this size. Residuals and fitted values are already stored, so sample
# 5000 points from those for the diagnostic plots.
set.seed(1)
idx <- sample(seq_len(nrow(airport)), min(5000, nrow(airport)))
fit <- fitted(model2)[idx]
res <- resid(model2)[idx]

par(mfrow = c(1, 2))
plot(fit, res, pch = ".", col = rgb(0, 0, 0, 0.3),
     xlab = "Fitted", ylab = "Residual", main = "Residuals vs fitted")
abline(h = 0, col = "red")
qqnorm(res, pch = ".", main = "Normal Q-Q")
qqline(res, col = "red")
```



```
par(mfrow = c(1, 1))
```

The residual-vs-fitted plot is where the right skew shows up: if the spread of residuals fans out as fitted values grow, that is heteroskedasticity, and it is the reason I refit on the log scale below and report those coefficients as percent changes. The Q-Q plot checks whether the residuals are roughly normal. Heavy tails there point in the same direction.

4.5 Log-scale robustness check

```
model2_log <- feols(log(FLT_TOT_1 + 1) ~ MONTH_MON + YEAR | APT_ICAO, data = airport)

tibble(
  r.squared      = get_r2(model2_log),
  within.r.squared = fixest::r2(model2_log, type = "wr2"),
  nobs           = nobs(model2_log)
) |>
  kable(digits = 3,
        caption = "Model 2 refit on log(FLT_TOT_1 + 1). Coefficients are approximate proportional change")
```

Table 8: Model 2 refit on $\log(\text{FLT_TOT}_1 + 1)$. Coefficients are approximate proportional changes.

r.squared	within.r.squared	nobs
0.93	0.141	125771

```
tidy_coefs(model2_log) |>
  filter(str_starts(term, "MONTH_MON") | term == "YEAR2026") |>
  mutate(pct_change = (exp(estimate) - 1) * 100) |>
  kable(digits = 3, caption = "Log-scale month and year coefficients, with implied percent change.")
```

Table 9: Log-scale month and year coefficients, with implied percent change.

term	estimate	std.error	statistic	p.value	pct_change
MONTH_MONAPR	0.269	0.015	17.484	0.000	30.869
MONTH_MONMAY	0.341	0.022	15.423	0.000	40.675
MONTH_MONJUL	0.444	0.029	15.287	0.000	55.858
MONTH_MONAUG	0.421	0.030	14.194	0.000	52.412
MONTH_MONSEP	0.428	0.026	16.254	0.000	53.348
MONTH_MONDEC	0.018	0.011	1.640	0.102	1.774
MONTH_MONFEB	0.084	0.008	10.661	0.000	8.728
MONTH_MONJUN	0.440	0.027	16.231	0.000	55.235
MONTH_MONMAR	0.118	0.008	13.934	0.000	12.531
MONTH_MONNOV	0.081	0.013	6.044	0.000	8.425
MONTH_MONOCT	0.334	0.021	15.969	0.000	39.649
YEAR2026	-0.023	0.010	-2.400	0.017	-2.314

On the log scale a coefficient of 0.10 for July means daily traffic at an average airport runs roughly 10% higher in July than in January. I report this version in the executive summary and slides. The log form drops the assumption of equal additive effects across airports of widely different sizes.

5. Interpretation in domain terms

The two models point to one conclusion. Daily flight volume rises and falls with the season. Summer months sit well above the January baseline at the average airport. The country mix in the sample does not drive this shape. Month coefficients barely move when airport fixed effects enter the model. July adds about 52 flights in Model 1 and about 54 in Model 2, near-identical figures.

The country effect behaves the opposite way. Model 1 ranks Denmark and Finland near the top. The country table explains why. Denmark contributes one airport, Copenhagen. Finland contributes one airport, Helsinki. Their high averages describe two busy hubs, not two busy countries. Once airport enters the model, those country signals fold into the airport fixed effects and lose meaning.

The year coefficient compares 2026 to 2025 with month and airport held constant. The effect is small and sensitive to scale. In the level model, 2026 runs about 2 flights per day above 2025. In the log model, 2026 runs about 2 percent below. The two specifications split on sign and agree on size, close to zero. In practical terms, 2025 and 2026 look alike.

6. Limitations

Four limitations shape the conclusions.

The geographic scope is narrow. The source is EUROCONTROL coverage. The findings describe European IFR traffic, not global aviation. North American, East Asian, and African airports sit outside the sample.

The outcome mixes flight types. Daily total movements combine passenger, cargo, business, and general aviation. A 200-movement day at a cargo hub and a 200-movement day at a tourist airport count the same. Separating them needs data the EUROCONTROL daily extract does not provide.

Two years limits the seasonal claim. The pattern repeats across 2025 and 2026, which rules out a one-year fluke. Two years still does not separate season from year-specific shocks. A strike, a volcanic closure, or a geopolitical event in one summer would enter the model as a month or year effect with no warning sign.

The design is observational. Airport fixed effects remove airport-level confounding, not all confounding. The data supports one claim. July runs busier than January at the average airport. The data does not support a causal claim about summer demand.

7. What I would do next

Three extensions would strengthen the work.

Split the outcome into departures and arrivals. This would show whether tourism creates timing asymmetries, more arrivals in early summer and more departures in late summer at sun destinations.

Add country-level predictors. GDP per capita, tourist-arrival share, and public-holiday calendars would give the country comparison real content. The current model drops country because airport absorbs the country signal, so these predictors would need a design without airport fixed effects.

Fit a mixed-effects model with random airport intercepts. This would return the same airport-controlled month estimates and report how much unexplained variance sits at the airport level versus the day level.